The Paradox of Equality: Redistribution in a Malthusian Economy

A Malthusian economy, described by Thomas Malthus, suggests that population growth offsets improvements in living standards because of diminishing returns to labor and scarcity of resources. Ashraf and Galor (2011, p. 2004) explain in their paper *Dynamics and Stagnation in the Malthusian Epoch* that a preindustrial economy was defined by the "counterbalancing effect of population growth on the expansion of resources." This essay argues that cutting the extraction of resources from the elites in a Malthusian economy only improves workers' life for a short period of time, but in the long run, the population growth offsets those temporary benefits. Even if there is rebellion that causes an increase in population and a greater total economic output, diminishing returns to labor reduce output per worker, causing that workers' consumption and utility return to their original equilibrium (same as before the rebellion).

I simulated a model based on the Malthusian economy metrics proposed by Ashraf and Galor (2011). My simulations compare two different scenarios. In the first one, a small group of elites, which we will call the "Lords", appropriate half of the total production; in the second one, the workers rebel and get to keep their whole production. Subsequently, I incorporated stochastic and Monte Carlo simulations to the model to explore how agricultural uncertainty and population growth influences the transition to the new equilibrium and the duration of the newly gained benefits in workers.

The model uses Cobb-Douglas production function ($Y_t = AL_t^{\alpha}H^{(1-\alpha)}$) to analyze the initial equilibrium, in which A is technology, L is workers population and H is the fixed land. Because of diminishing returns to labor, population growth reduces output per worker. Since the Lords take 50% of the production, workers only keep $w_t = (1 - \tau)(Y_t/L_t)$. According to Ashraf and Galor (2011), populations reach equilibrium once every adult has on average one surviving child, keeping population constant. As a result, workers' retained income settles at $w^* = \rho/\gamma$, in which ρ is the cost of raising a surviving child and γ is the fraction of income allocated to taking care of the child.

When workers rebel and take the Lords out of the equation, extraction goes from 50% to 0%. Since population, land, and technology initially remain the same, total output and output per worker stays steady. As figure 1 displays, initially, population, technology and land remain constant. However, workers now retain the totality of their production, doubling their retained income, consumption and utility. As a result, the expected value of surviving children per adult also increases, leading to population growth.

In the long run, the extra surviving children makes the population grow. But since land is fixed, diminishing marginal returns on work reduces the output per worker. As Table 1 shows, the population grows from 13,801 to 100,000, while total production increases from 27,602 to 100,000. As a result, output per worker reduces by half. At the end, utility and consumption go back to their original values, despite the fact that the new economy has a new equilibrium with higher population and higher total output.

To check if these results stay the same under circumstances with uncertainty, I did 5000 Monte Carlo simulations with random variations in both agricultural productivity and reproduction. Figure 2 and Table 2 show that the initial benefits of the revolution do not last long. After 5 generations, the average gain in consumption is approximately 8.37%; after 10 generations the average gain is approximately 0.94%; and after 50 generations, the average difference is practically zero. The results of the Monte Carlo simulations suggest that the prediction of the Malthusian model still holds under the proposed assumptions, even when we add randomness and uncertainty to the model.

This model lets us explore the relationship between inequality and other economic variables. As Figure 3 illustrates, when the Lords take a larger portion of the outcome, the equilibrium population's and total production in equilibrium decreases. However, output per worker increases because there are less workers in the same amount of land. Interestingly, worker consumption remains constant in long term equilibrium, regardless of how much output is taken by the Lords. These results suggest that greater inequality resulting from elite extraction could be associated with a smaller equilibrium population, without necessarily changing workers' long run consumption.

These predictions could be compared using historical data of regions that had different amounts of extraction from the elites. For example, we could compare data from taxes and feudal rents with population data, agricultural production and real salaries. A way of doing this would be to analyze regions in which the extraction decreased and compare it to other regions in which extraction remained constant, observing how population and live conditions changed across time. Nonetheless, establishing a causal relationship is not an easy task, as different societies were affected by external factors such as war, technological advancements and institutional transformations.

Appendix

Figure 1. Economic Adjustment Following the Elimination of Lords

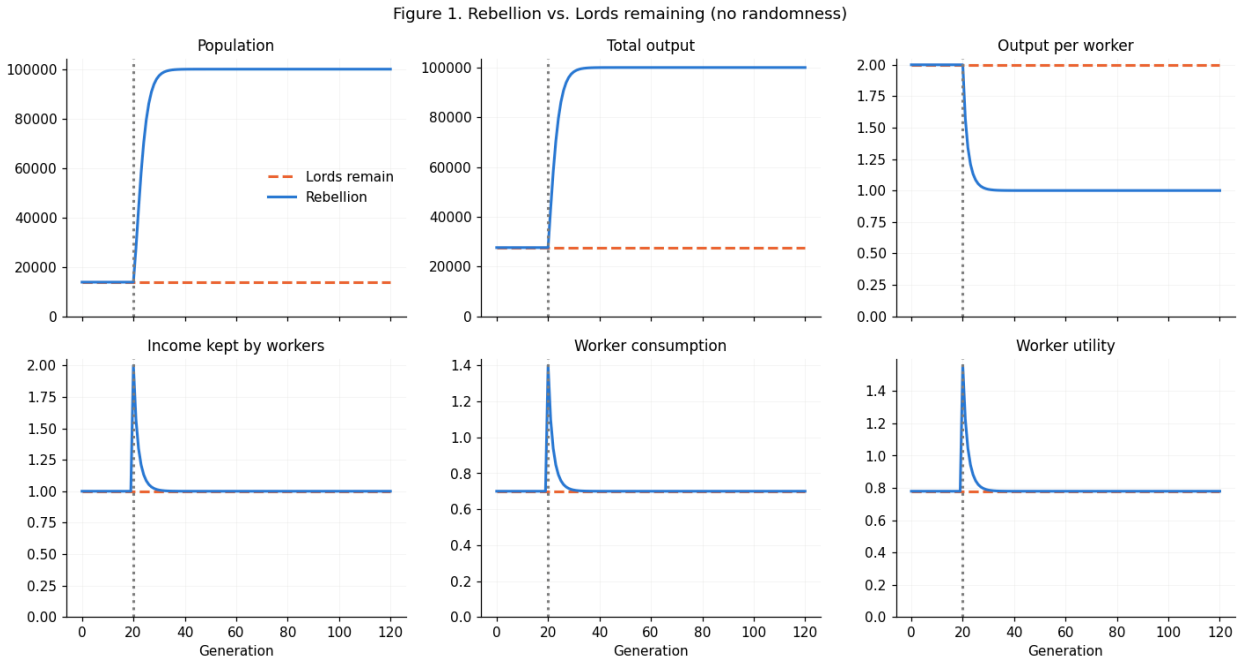


Figure 2. Monte Carlo Effects of the Rebellion

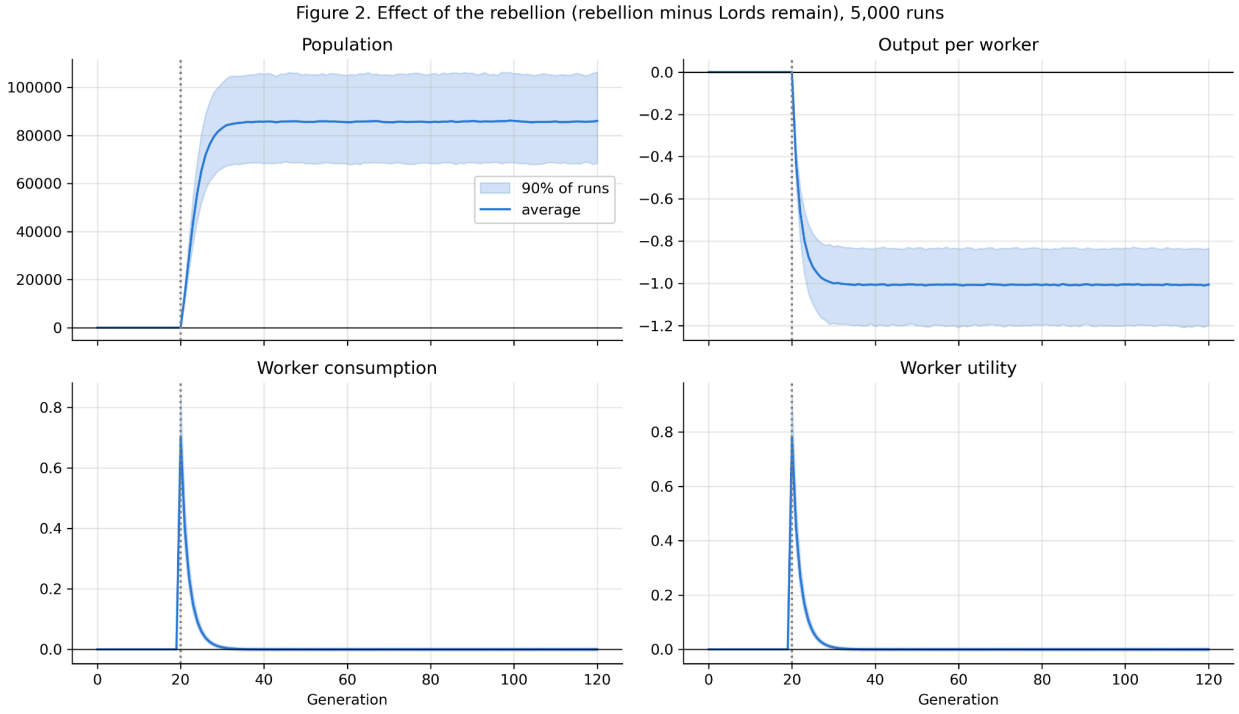


Figure 3. Elite Extraction and Long-Run Malthusian Equilibrium

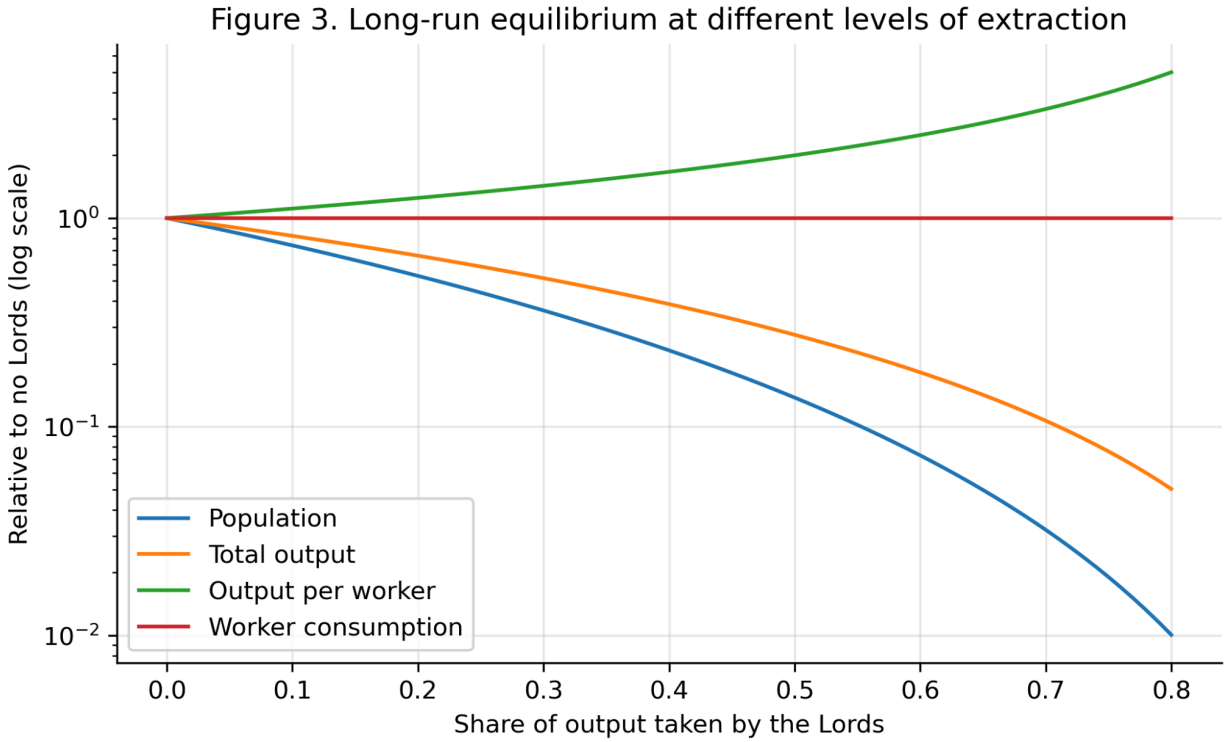


Table 1. Malthusian Equilibrium Before and After Rebellion

	With Lords	Without Lords	Ratio
population	13,801.119	100,000	7.246
output	27,602.238	100,000	3.623
output_per_worker	2	1	0.5
income	1	1	1
consumption	0.7	0.7	1
utility	0.779	0.779	1

Table 2. Monte Carlo Simulation Results

	average gain	95% CI	share of runs better off
0	100.00%	100.00% to 100.00%	1.0
5	8.37%	8.36% to 8.38%	1.0
10	0.94%	0.93% to 0.95%	0.9856
25	0.01%	-0.00% to 0.02%	0.5172
50	0.00%	-0.01% to 0.01%	0.4992

References

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